

Demo Guide — Clinical Research Assistant

Presenter playbook + functional-test checklist. Organised around the **six clinical-research lifecycle phases** so a demo can walk a researcher / sponsor / institution through their actual workflow — pick the phase that matches your audience and run the demos in order, or cherry-pick individual demos for a focused conversation.

Every paste target lives in a fenced code block — click the copy icon, drop it into the chat input, move on. Use this same guide for end-to-end functional testing: each demo carries **sanity checks** that double as an acceptance test, plus **failure modes** that flag what to look for.

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***Naming.** Each demo is identified by <Phase letter><number> — E■PH1■ for trial Design, S■PH2■ for eXecution, A■PH3■ for Cross-cutting. Lets a reviewer reference "X4 AE/SAE" without ambiguity.*

Before you start

The app is deployed on AWS and reached over HTTPS through a load balancer — there's no local stack to bring up. Just open the URL and sign in.

App URL: `https://cra-test-alb-1950107331.us-east-1.elb.amazonaws.com/`

First-time access

- The site uses a **self-signed TLS certificate**, so your browser shows a security warning on first visit. Accept it once (**Advanced** → **proceed to site**) — do this *before* signing in, or the login redirect can be blocked on the certificate error.
- Access is gated by a **Cognito login at the load balancer**: you're redirected to a hosted sign-in page, then back to the app. You need a Cognito account in the pool — ask the admin for an invite if you don't have one.
- After signing in, **hard-refresh** (Ctrl+F5) so the sidebar bell, watches / multisite / portfolio links, and the latest specialist cards load.

Surface map (all the URLs you'll touch)

`https://cra-test-alb-1950107331.us-east-1.elb.amazonaws.com/`

`https://cra-test-alb-1950107331.us-east-1.elb.amazonaws.com/admin.html`

• `validation-pack`

– main app (chat workflows)

– paper-source admin · role admin

https://cra-test-alb-1950107331.us-east-1.elb.amazonaws.com/sr.html	– SR screening (Phase 01)
https://cra-test-alb-1950107331.us-east-1.elb.amazonaws.com/watches.html	– watches + group subscriptions (Phase 01)
https://cra-test-alb-1950107331.us-east-1.elb.amazonaws.com/ecrf.html	– Form Builder (Phase 04, admin)
https://cra-test-alb-1950107331.us-east-1.elb.amazonaws.com/collector.html	– Site EDC + safety + recruitment + visit + drug + labs (Phase 04)
https://cra-test-alb-1950107331.us-east-1.elb.amazonaws.com/epro.html?token=...	– Participant ePRO (Phase 04, magic-link)
https://cra-test-alb-1950107331.us-east-1.elb.amazonaws.com/multisite.html	– Central-coordinator multi-site rollup (Phase 04)
https://cra-test-alb-1950107331.us-east-1.elb.amazonaws.com/portfolio.html	– Portfolio + budget dashboard (Phase 06)
https://cra-test-alb-1950107331.us-east-1.elb.amazonaws.com/library.html	– Library + RAG (Phase 06)

Pre-seed data for Phase-04 + Phase-05 demos (the "Smoke deployment")

Almost every Execution-phase + Analysis-phase demo wants an existing deployment with a site + at least one subject. Run **X1** → **X2** once first to produce the SMOKE-T2DM-001 study + Smoke deployment + Site 01 + S01-001 subject. Subsequent Execution + Analysis demos assume that exists.

AWS quota note

The watches demo cadence (■PH0■ ■PH1■ *) issues ~30 Bedrock calls/hour. Switch any demo watches to a weekly schedule when you're done, or delete them — otherwise they'll keep firing.

Phase 01 — Evidence synthesis

***Phase context.** Before any new study runs, the team needs to know what the existing evidence base looks like — for the protocol's background section, for the IRB packet's "what's known", for the grant application's justification of need, and for guideline + HTA submissions whose entire artefact is a synthesis.*

E1 — Meta-analysis on synthetic data (≈ 5 min)

Demos. End-to-end meta-analysis with anti-hallucination guardrails: PICO → search → extraction → forest plot, all inside the sandbox. Most reliable demo path because the data is deterministic.

Goal. 8-row forest plot, pooled RR ≈ **0.44** (95% CI ~ 0.33–0.58), I² in the **30–50%** range.

*The numbers below are **fabricated** to give a plausible meta-analysis result. Author names are made up so this can never be confused with real evidence.*

Steps

1. New conversation. Paste the canonical research question:

Does adding a proton pump inhibitor to dual antiplatelet therapy reduce upper GI bleeding in patients who recently underwent PCI for acute coronary syndrome?

2. At the PICO confirmation step, paste:

Element	Value
---	---
Population	Adults post-PCI for ACS, on dual antiplatelet therapy (aspirin + P2Y12 inhibitor)
Intervention	PPI added to DAPT (any PPI, any dose, ≥30 days)
Comparator	DAPT alone or DAPT + placebo
Outcome (primary)	Upper GI bleeding events during follow-up
Study design	Randomized controlled trials

3. Skip / fast-forward the search step. (You can let it run on real PubMed — the synthetic studies won't be there, that's fine.)

4. At the extraction step, paste:

Study	Year	Design	PPI events	PPI total	Control events	Control total	Follow-up (mo)
Anderson et al.	2014	Multicenter RCT	12	1200	28	1198	12
Bekele et al.	2016	Single-center RCT	4	320	9	318	6
Chen et al.	2018	Multicenter RCT	18	2200	41	2180	12
Diaz et al.	2019	Pragmatic RCT	7	850	14	845	9
Eriksen et al.	2020	RCT	5	560	7	562	6
Faruq et al.	2021	Multicenter RCT	10	1400	25	1395	12
Garcia et al.	2022	RCT	3	480	8	478	6
Huang et al.	2023	Multicenter RCT	14	1800	33	1810	12

(Totals: PPI 73 / 8810 (0.83%) vs Control 165 / 8786 (1.88%).)

5. On the Data Extraction card, click the primary button labeled "Run meta-analysis (N/N rows complete)".

This button — *not* a free-form message — emits the magic prefix `Confirmed extracted data ...` plus the extraction JSON, which is what triggers the model to call `sandbox_exec` and render the forest plot.

■ **Don't type "approved", "go ahead", "run it" etc. in the chat input.** The thread will stay on the meta-analysis specialist (it's pinned), but without the JSON payload the model can't call `sandbox_exec` — it will compute the pooled numbers inline from conversation history and emit a result with `forest_plot_image: null`. The runtime validator (`_require_sandbox_for_results`) will force a retry, but you'll burn an extra Bedrock round-trip per attempt. Always click the button.

6. On the Meta-Analysis Results card, click "Download PDF" or "Download DOCX" to grab a manuscript-style report.

The report bundles, in this order: research question · PICO table · included studies table · per-outcome pooled estimate (RR, 95% CI, I^2 , heterogeneity p) · embedded forest plot · clinical interpretation · summary · caveats.

7. From the same terminal card, click "→ Draft as manuscript", "→ Draft GRADE", or "→ Draft lay summary" to demonstrate the cross-handoff into Phase 05 specialists (each seeds a new thread with the meta-analysis JSON verbatim).

Sanity checks

- Pooled RR $\approx$ **0.44** (95% CI roughly 0.33–0.58), random-effects.
- I^2 in the **30–50%** range — some between-study variation, not extreme.
- 8 study rows + diamond. Plot well under sandbox limits (figsize / dpi / 2 MB).
- **Funnel plot + Egger's test** appear automatically in STEP 5 (beyond-forest-plot viz, see A2).
- "Take it away" panel offers PDF + DOCX; the three handoff buttons (manuscript / GRADE / lay summary) seed new threads.

Variant — stress-test heterogeneity (optional)

Swap the step-4 extraction with this table to force $I^2 \approx$ **75–85%** and a wide pooled CI. Two outliers (Diaz 2019, Faruq 2021) favor control.

Study	Year	Design	PPI events	PPI total	Control events	Control total	Follow-up (mo)	Approx RR
Anderson et al.	2014	Multicenter RCT	8	1200	20	1198	12	0.40
Bekele et al.	2016	Single-center RCT	3	400	12	395	6	0.25
Chen et al.	2018	Multicenter RCT	15	1500	25	1495	12	0.60
Diaz et al.	2019	Pragmatic RCT	22	600	12	605	9	1.85

Eriksen et al.	2020	RCT	4	700	16	695	6	0.25	
Faruq et al.	2021	Multicenter RCT	18	900	14	895	12	1.28	
Garcia et al.	2022	RCT	5	800	19	795	6	0.26	
Huang et al.	2023	Multicenter RCT	11	1300	24	1305	12	0.46	

E2 — Real evidence pipeline: Protocol → Search → Meta → RoB (≈ 15 min)

Demos. Coherence across handoffs — PICO carries through every specialist, so the protocol's eligibility criteria match the search strategy, which matches the meta-analysis inclusion set, which matches the RoB cohort.

E2a. Draft a PRISMA-P protocol

Trigger (either):

/protocol SGLT2 inhibitors for HF prevention in T2DM

Help me draft a PRISMA-P protocol on whether SGLT2 inhibitors reduce heart-failure hospitalization in adults with type 2 diabetes.

Expected flow: protocol_methods card → edit fields → click **Methods confirmed — draft full protocol** → protocol_document (draft) with grounded background + numbered citations + PROSPERO field map → free-form refinements or **Finalize protocol** → final card with **Build the search strategy** + **Skip to meta-analysis** handoff buttons + **Download PDF / DOCX**.

E2b. Build the search strategy

From E2a, click **Build the search strategy**. Or standalone:

/search proton pump inhibitor + DAPT in post-PCI ACS

Expected flow: query_blocks card with MeSH chips → edit synonyms / drop unwanted MeSH → **Confirm terms** → strategy_result (draft) with composed Boolean + live hit counts → click refinement buttons until band-status banner turns green → **Finalize strategy** → adds **Run meta-analysis on these results** + **Watch this query** buttons.

E2c. Hand off to meta-analysis

Click **Run meta-analysis on these results**. New thread arrives at PICO step already understood; STEP 3's search query is the validated string. For a deterministic forest plot, drop in E1's synthetic extraction at this point.

E2d. Assess risk of bias

From the meta-analysis data_extraction card, click **Assess risk of bias** — pre-loaded with the extracted studies. Standalone:

/rob Run RoB 2.0 on PMID 20925534, PMID 30873575, PMID 31091374

Expected flow: rob_assessments card with 5 (RoB 2.0) or 7 (ROBINS-I) domains → edit judgments → **RoB confirmed — generate summary** → stacked-bar plot + per-domain count table → **Finalize RoB** → final card with **Re-run meta-analysis excluding high-RoB studies** + **Download PDF / DOCX**.

Sanity checks

- **RoB auto-suggest.** study_designs = ["Randomized Controlled Trial"] only → defaults to RoB 2.0; add "Cohort study" → switches to ROBINS-I.
- **Citation provenance.** Every reference card has an origin badge: search_papers (cyan), web_search (blue), wikipedia (amber). PubMed refs MUST display a PMID; web sources MUST display a URL.
- **No fabricated numbers.** Background should not contain percentages, sample sizes, or effect estimates that aren't tied to a [N] citation.

- **PROSPERO placeholders are honest.** User-specific fields (Named contact, Funding sources, IRB number) display content starting with [USER INPUT NEEDED: styled in amber.
- **Quotes are verbatim.** RoB italic blue quote blocks must appear word-for-word in the linked abstract.

E3 — SR title/abstract + full-text screening UI (≈ 10 min)

Demos. The dual-review screening surface that bridges search (which returns thousands of hits) and meta-analysis (which uses the dozen that meet criteria). Closes the single biggest time-sink in any SR.

Pre-req. Researcher role (carries `sr.create`).

Open at: <https://cra-test-alb-1950107331.us-east-1.elb.amazonaws.com/sr.html>

Steps

1. Create a SR project. Click + **New project**. Fill in:

- **Title:** SGLT2 inhibitors for HF prevention in T2DM (smoke review)
- **PICO:** Adults with T2DM · SGLT2 inhibitor · standard care · HF hospitalisation.
- **Inclusion criteria:** RCT, English, ≥12 weeks follow-up, HbA1c reported at baseline.
- **Exclusion criteria:** Type 1 diabetes; pre-existing HF NYHA III-IV; no quantitative outcome data.

2. Add reviewers. From the project page → **Manage reviewers**:

- Invite a researcher by email → grant role `reviewer_1` at `scope_type='sr_review'`, `scope_id=<project_id>`.
- Invite a second researcher → grant role `reviewer_2`.
- Invite a third → grant role `adjudicator`.

3. Ingest candidates. Click **Run search** → reuses the multi-source paper search to populate `SrCandidate` rows. For a deterministic demo, paste a small canned list:

```
PMID 30859901
PMID 31535827
PMID 32865375
PMID 32865377
PMID 33069326
```

4. Walk through the screening queue. Open the candidate queue. For each abstract:

- **AI-assist suggestion** appears inline (include / exclude / maybe + confidence) — pre-classification by the `sr_screening_assist` specialist.
- Press keyboard shortcut: `i` = include · `e` = exclude · `m` = maybe.
- Reviewer 1's decisions are blind to Reviewer 2's — switch user accounts to demo the parallel review.

5. Adjudicate conflicts. Once both reviewers complete their pass, conflicts surface in the adjudicator's queue. Adjudicator clicks include / exclude / kick-back-for-discussion.

6. Generate PRISMA flow. From the project page → **PRISMA flow** → hand-rolled SVG diagram opens at `GET /api/sr/projects/{id}/prisma/diagram.svg`. The standard 4-row flow with side exclusion boxes including per-reason breakdown.

Sanity checks

- **Dual review is blind.** Reviewer 1's response shape carries no Reviewer 2 votes; switching to Reviewer 2's account shows their own queue with no agreement indicator until adjudication.
- **AI-assist persistence.** AI suggestions land in `AI Suggestion` rows so accuracy-vs-human can be measured later. Override behaviour: reviewer's manual decision is recorded separately from the AI suggestion (both survive in the audit trail).
- **Agreement rules.** Both include → terminal `include`; both exclude → terminal `exclude`; disagree → `pending_adjudication`.
- **PRISMA counts derive entirely from `SrCandidate.status` + reason codes** — no fabrication.

Failure modes

- Missing `sr_review` scope grant → user can't see the project at all (404, not 403, to avoid id leaks).
- Adjudicator without conflicts visible: confirm both reviewers have committed their pass.

E4 — Network meta-analysis (NMA) (≈ 8 min)

Demos. When the clinical question involves ≥ 3 interventions (e.g. "of the five direct oral anticoagulants, which has the best safety-efficacy balance?"), pairwise meta-analysis can't answer it. NMA pools direct + indirect evidence across the network.

Trigger:

`/nma compare five direct oral anticoagulants for stroke prevention in non-valvular AF – apixaban, dabigatran, edoxaban, rivaroxaban, warfarin`

Expected flow: Five-stage workflow — intake → `PicoNetwork` with ≥ 3 interventions + **transitivity rationale** (schema-required) → search → per-arm extraction → NMA results with league table + SUCRA + network-geometry PNG. Default backend is **frequentist** (`mvmeta` + electrical-network analogy on `numpy` + `scipy`; 1000 MVN posterior draws). Type Run `Bayesian NMA` after the intake step to opt into the PyMC NUTS backend.

Sanity checks

- **Transitivity rationale required.** Schema rejects an NMA intake without `pico.rationale` — the operator must justify why indirect comparisons are valid (similar populations, comparable outcome definitions).
- **League table is colour-coded** square matrix — green = protective, red = risk-increasing, white = null-crossing.
- **SUCRA ranking** sums to ~ 1.0 across interventions.
- **Network geometry PNG** renders with node size $\propto \sqrt{n_{\text{studies}}}$, edge width $\propto n_{\text{head_to_head_trials}}$.
- **Anti-hallucination.** PMIDs from `search_papers` only; pooled effects + CIs + SUCRA from sandbox runs only.
- **Bayesian backend** writes `skip_reason` when PyMC is missing from the sandbox image (documented deploy-time gate).

Failure modes

- `/nma` with only 2 interventions → schema error from `PicoNetwork.interventions min_length=3`.
- Bayesian backend unavailable → `skip_reason="PyMC not installed in sandbox image; rebuild sandbox with pymc>=5"`.

E5 — IPD meta-analysis (≈ 8 min)

Demos. When you have subject-level data from included trials (not just published summaries), IPD MA is the gold standard. One-stage + two-stage run side-by-side as the methodological diagnostic.

Trigger:

/ipd individual patient data meta-analysis on statins for primary prevention of cardiovascular events

Expected flow: 5-stage workflow — intake → bundle with per-trial CSV pastes + column mapping → main results (one-stage + two-stage side-by-side) → subgroup × treatment interaction → assembled document.

Paste a minimal canned bundle when the bundle step opens:

```
Trial: ASCOT-LLA
subject_id,treatment,outcome,age,sex
S001,1,0,55,M
S002,1,1,62,M
S003,0,1,58,F
S004,0,0,49,M
```

```
Trial: WOSCOPS
subject_id,treatment,outcome,age,sex
W001,1,0,60,M
W002,0,1,67,M
W003,1,0,54,F
W004,0,0,52,F
```

Sanity checks

- **One-stage AND two-stage always run side-by-side.** A divergence > 0.2 in log-scale flags model misspecification.
- **Sandbox-backed.** `one_stage.py` runs statsmodels MixedLM (continuous) / GLM Binomial logit / stratified PHReg (TTE); `two_stage.py` does per-trial estimate + DerSimonian-Laird random-effects pool with $I^2 + \tau^2$.
- **Anti-hallucination.** Per-trial estimates + $I^2 + \tau^2$ + interaction p come from sandbox runs only.

E6 — GRADE Summary of Findings + PRISMA 2020 checklist (≈ 5 min)

Demos. Every journal-acceptable systematic review now requires a GRADE SoF + PRISMA 2020 reporting checklist. Most teams produce these as the very last step before manuscript submission.

Pre-req. A completed E1 or E2 meta-analysis in the same thread (or use the cross-handoff from the meta_analysis terminal card).

Trigger (either):

/grade Generate a GRADE SoF for the PPI/DAPT meta-analysis

Or click → **Draft GRADE** on the meta-analysis terminal card.

Expected flow: 5-stage workflow — intake → per-outcome assessment → SoF assembly → PRISMA 2020 checklist → assembled document. Certainty is COMPUTED via Pydantic `computed_field` from per-domain ratings (RCT start=4 / observational=2; serious=-1, very_serious=-2; observational upgrades for large effect / dose-response / residual confounding).

Sanity checks

- **Certainty cannot be inline-asserted.** Try to override `certainty` in a turn → schema rejects (it's a `computed_field`).

- **Every downgrade rationale must cite a source number** (I^2 value, CI bounds, n_{studies} , Egger's p). System prompt forbids handwaving.
- **PRISMA 2020 checklist** carries 42 canonical sub-items from Page et al. (BMJ 2021). Unreported items render as "Not reported" — operator can't claim items they didn't fulfil.
- **Visual SoF chip table** auto-rendered host-side from assessments (no agent involvement) — colour-coded green / amber / red per the GRADE-pro convention.
- Landscape PDF + DOCX export.

E7 — Living-review watch (≈ 5 min setup, then background)

Demos. Scheduled re-search + materiality-alert loop. Setup is ~5 minutes; the rest is background.

Architecture in one paragraph. `POST /api/watches` runs the saved search once to populate the baseline PMID set, persists the watch, and registers an APScheduler cron job. At each fire, `services/watch_runner.py` re-executes the search via `tools/clinical/search_papers._fan_out`, computes the diff against `baseline_pmids.json`, hands new papers to `agent/specialists/watch_triage.py`, persists a `WatchRun`, and creates a `Notification` if `notify=true`. The bell polls `/api/notifications/unread-count` every 30 s.

Steps

1. Build a search strategy end-to-end (E2b) to a finalized `strategy_result`.
2. On that card, click **Watch this query**.
3. In the modal: - **Name** — pre-filled from the original research question. - **Schedule preset** — pick **Every 2 minutes (demo)** to see the loop without waiting. - **Materiality threshold** — leave at 0.6. - Click **Create watch**.
4. Click sidebar **■ Watches** → the new watch appears with status `active`, baseline = 5 PMIDs, cron **■PH1■ ■PH2■ ***, next-run timestamp ~2 minutes out.
5. Wait 2 minutes (or click **Run now**). Click **Show runs** to expand history.
6. If success AND any triage hit `materiality ≥ 0.6`, the bell badge lights up. Click → dropdown → click a notification → marks read + jumps to `/watches.html`.

Sanity checks

- **Baseline pre-populated.** Watch-creation logs show pre-populated baseline with N PMIDs from initial search.
- **Triage pills are honest.** `relevance=high + design_fit=matches + materiality > 0.6` reserved for clearly in-scope, well-powered studies.
- **Restart durability.** Restart agent container → startup log shows Scheduler re-registered N active watch(es) from DB.

Failure modes

- **AWS quota.** **■PH1■ ■PH2■ *** = 30 Bedrock calls/hour. Switch to weekly after the demo.
- **Bad cron.** Returns 400 with parser error.
- **All sources disabled in `/admin.html`** → `status=error` with "no sources enabled".

E8 — Group-level living-review subscriptions (quorum) (≈ 5 min)

Demos. Wrap an existing watch with an invited panel that votes yes / no / abstain on each new run. Closes the "guideline committee" gap that personal watches couldn't fill — HTA bodies need consensus, not per-individual alerts.

Pre-req. A finalised watch from E7.

Steps

1. Open `/watches.html` → scroll to the new "Group subscriptions" section (at the top).

2. Click "+ New subscription". Fill in:

- **Name:** Cardio guideline panel
- **Watch:** select the E7 watch from the dropdown.
- **Min votes:** 2
- **Min fraction:** 0.5

Click **Create**. The subscription card lists "0 voters · 0 observers · $\geq \max(2, 50\%)$ ".

3. **Add members.** Click **Manage members** → paste an email + select role (voter / observer) → **Add**. Repeat for 2-3 voters + 1 observer.

4. **Trigger a watch run** (either click **Run now** on the underlying watch, or wait for the cron to fire).

5. **Cast votes.** From the subscription card → **Show runs + votes** → for the newest run, click **Vote YES** (optionally type a rationale prompt). Switch user accounts and cast votes from other voters until quorum clears.

6. **Verify group notification fan-out.** Once quorum clears, the bell badge increments by `n_members` for every member's account. Each member's notification feed shows a [Group quorum] titled alert with the run id + paper count.

Sanity checks

- **Observers don't count toward the denominator AND can't vote.** Try to cast a vote from an observer account → 422 with "Only voting members may cast votes...".
- **Vote idempotency.** Re-casting from the same voter overwrites rather than appending (UNIQUE constraint on (subscription, run, voter)).
- **Paused subscriptions reject votes outright.** Switch the subscription to `paused` → vote attempt returns 422 with "Subscription is 'paused'...".
- **Idempotent per-recipient fan-out.** A stray double-tally doesn't double-notify; existence check before insert.

Failure modes

- **Cross-watch run rejected.** A run from a different watch's id returns 422 ("Run X not found on this subscription's watch").
- Empty subscription (0 voters): quorum never clears regardless of threshold.

Phase 02 — Trial design

***Phase context.** Once the evidence base supports a new trial, the team specifies the question (PICOT), powers it (sample size), and writes the analysis plan. Deliverables: protocol body + Statistical Analysis Plan,*

often submitted to peer review before any subject enrolls.

D1 — Sample-size + SAP drafter (≈ 10 min)

Demos. Four-stage workflow producing an ICH-E9-shaped Statistical Analysis Plan with downloadable PDF + DOCX. Sample-size calculator runs directly on `scipy.stats` — no sandbox round-trip.

Trigger:

/sap two-arm RCT of CBT vs SSRI for depression, primary outcome HAM-D at 12 weeks

Expected flow:

1. **PICOT intake** card — Population, Intervention, Comparator, Outcome (with measurement scale), Time. Edit fields → **PICOT confirmed**.
2. **Sample-size derivation** — pick formula (two-proportions / two-means / time-to-event Schoenfeld / paired); fill effect size + α + $1-\beta$ + dropout. The card calls `sample_size` tool host-side and returns N per arm + total + dropout-inflated N. **Sample size confirmed**.
3. **Analysis plan** — ICH-E9-shaped sections (study objectives + estimands + analysis populations + primary + secondary analyses + interim + sensitivity + missing-data handling). **Analysis plan confirmed**.
4. **Assembled SAP document** — full Markdown + PDF + DOCX download.

Sanity checks

- **Sample-size return shape matches** ``SampleSizeResult`` — `N_per_arm`, `total_n`, `total_inflated_n`, formula, assumptions echoed back.
- **Four formulas supported**. Each one rejects nonsensical inputs (negative effect size, α outside (0,1)).
- **PDF carries the full SAP** — Title · PICOT · Sample size · Analysis plan · Methods · Assumptions · Plus statistical sensitivity tables when present.
- **Estimands framework** required (ICH E9(R1)). Each primary analysis has an `estimand` block (target population + treatment + endpoint + ICE handling + summary measure).

Variant — non-inferiority (currently deferred but worth narrating)

The four superiority formulas ship; non-inferiority + equivalence are roadmap items. Surface the limitation: "today the calculator covers superiority; NI / equivalence formulas are a follow-up."

Phase 03 — Start-up

***Phase context.** The trial gets registered with regulators and the ethics committee (IRB / IRC) approves the protocol + ICF. Without these the trial cannot enrol its first subject. Both deliverables have strict structured templates and unforgiving compliance requirements.*

S1 — Trial-registration drafter (CT.gov + EU CTR / CTIS) (≈ 8 min)

Demos. Side-by-side drafts for ClinicalTrials.gov PRS and EU CTR / CTIS. NCT IDs and CTIS numbers are NEVER fabricated — assigned by registries on submission.

Trigger:

/register Phase 2 SGLT2i in HFpEF, double-blind randomised parallel, primary outcome KCCQ Total Symptom Score at week 24

Expected flow:

1. **Registration intake** — sponsor + indication + design + Phase + primary outcome + sites. Sponsor PHI fields marked [SPONSOR INPUT].
2. **Core fields** card — CT.gov + EU CTR field shapes side-by-side. Edit → **Core fields confirmed**.
3. **Registration drafts** — copy-paste-ready blocks for the CT.gov PRS portal and the EU CTIS portal, with NCT placeholder [CT.gov-assigned] + CTIS placeholder [EUDRACT-assigned]. **Drafts confirmed**.
4. **Assembled registration document** with background paragraph + PDF + DOCX export.

Sanity checks

- **NCT / CTIS IDs NEVER auto-filled.** Schema rejects anything matching NCT\d{8} from the model.
- **Background paragraph cites real evidence** via search_papers (PMIDs + origin badges).
- **Sponsor PHI sanitised** — [SPONSOR INPUT] placeholder colour-coded amber.

Cross-handoff

Click **Draft IRB packet from registration intake** on the final card → seeds an IRB drafter thread (see S2) with the registration's core fields pre-loaded.

S2 — IRB / ethics packet + ICF drafter (≈ 10 min)

Demos. Protocol synopsis (1-2 page IRB-triage summary) + Informed Consent Form (schema-enforced against 21 CFR §50.25(a)) + computed Flesch-Kincaid reading-level grade. Multilingual en/es/fr/de.

Trigger (or click the cross-handoff from S1):

/irb Phase 2 trial of Drug X in disease Y, US IRB, language English, reading grade 8

Expected flow:

1. **IRB intake** — protocol summary + jurisdiction + language + reading-level target + population descriptor. Default reading grade = 8.
2. **Protocol synopsis** — 8 ICH E6(R2)-aligned sections (design / objectives / endpoints / methods / statistics / eligibility / schedule / risks). **Synopsis confirmed**.
3. **Informed Consent Form** — schema-enforced 9-section minimum covering 21 CFR §50.25(a) elements A-I. ICF carries reading_level_target + reading_level_grade_actual (Flesch-Kincaid computed at draft time). Multilingual — try language: es to switch to Spanish.
4. **Assembled IRB packet** — synopsis + ICF + signature block + PDF + DOCX.

Sanity checks

- **ICF schema-enforced.** Try to emit an ICF with 8 sections → schema rejects (min_length=9).
- **Reading-grade overshoot flagged in PDF.** Above-target grades render with an amber warning band on the cover page.
- **Multilingual.** language: en|es|fr|de switches the ICF body language. Any other language → clarification asking for a human translator.
- **No medical-decision language.** Try to inject "you should consent" → system prompt blocks; ICF uses "you may choose to take part" framings.
- **No promissory benefit claims.** "you will get better" forbidden; "may benefit" / "may provide information that may help future patients" allowed.

Cross-handoff

Click **Draft lay summary from IRB packet** on the final card → seeds a lay-summary thread (see A6) with the protocol synopsis pre-loaded as the recruitment-language source.

S3 — Randomisation / IRT (≈ 8 min)

Demos. Schedule generation (DM role) → at-enrolment allocation (coordinator + PI roles) → emergency code-break (PI role only). Pulls TRT01P / TRT01A into SDTM ADSL on derivation.

Pre-req. SMOKE-T2DM-001 deployment from X1-X2.

Steps

1. Generate randomisation schedule. Switch to a data_manager account. Open the Smoke deployment's "Randomisation" panel in `collector.html`. Click **Generate schedule**:

```
{
  "blinding_mode": "double_blind",
  "algorithm": "permuted_block",
  "block_size": 4,
  "arms": [{"name": "Active", "ratio": 1}, {"name": "Placebo", "ratio": 1}],
  "n_subjects": 60,
  "random_seed": 12345
}
```

Schedule appears as a sequence of 60 allocations, masked (you see slot numbers, not arms, because `blinding_mode=double_blind`).

2. Allocate a subject at enrolment. Switch to a coordinator account. On the subject S01-001 page → **Randomize**:

POST `/api/edc/subjects/{subject_id}/randomize`

Returns `Allocation.id` + masked arm placeholder ("Arm A"). The allocation is bound to the next unused schedule entry.

3. Emergency code-break. Switch to the PI account. From the subject row → **Code-break**:

```
{
  "reason": "Subject presented to ED with severe hypoglycaemia; treating physician requires arm to choose appropriate management."
}
```

The reason is required (≥8 chars). The PI sees the actual arm; the audit trail records the code-break event with reason + actor + timestamp.

4. Re-derive CDISC ADSL to confirm TRT01P / TRT01A populated. Trigger:

POST `/api/edc/deployments/{deployment_id}/cdisc/derive`

→ Open the Submissions card → click **Download adsl.csv** → TRT01P column populated from `Allocation.arm` (not "TBD").

Sanity checks

- ``randomization.generate`` is **DM-only**. Coordinator allocation attempt to `/schedule` → 403.
- ``randomization.codebreak`` is **PI-only**. Coordinator + DM attempts → 403.
- **Code-break reason validated.** `reason < 8 chars` → 422.

- **Blinding mask honoured in API responses.** `GET /api/edc/subjects/{subject_id}/allocation` returns masked arm until `code-break flips unblinded=True`.
- **ADSL pulls from Allocation.** After derive, `TRT01P` matches the schedule's arm for each allocated subject; un-allocated subjects render `TRT01P="TBD"`.

Phase 04 — Execution

***Phase context.** Once the trial is approved and registered, real subjects enrol and real data accumulates. This is the largest, longest, and most operationally complex phase. The platform's regulatory-grade infrastructure (Part 11 / ALCOA+ / GCP / ICH E6) does the most work here.*

X0 — Trial-staff setup: invite, scope, onboard, countersign (≈ 15 min)

Demos. The regulatory-grade user-administration loop you walk before X1. Admin invites the trial team with scope-aware role grants; the SoD matrix blocks ICH E6 §5.18–5.19 + Part 11 §11.10(d) violations at invite time; each invitee completes an `/onboarding.html` wizard that captures the fields their role requires; clinical-data surfaces stay locked until onboarding is complete (server-side gate); the PI countersigns delegation entries with Part 11 §11.200 password reauth; every action lands in an append-only audit log.

Skip this demo for the smoke-deployment quick run — the legacy `default-user` placeholder is `admin` + everything in single-user dev mode. Run X0 when you want to walk through the actual multi-user posture an audit / inspection would see.

Pre-req. An admin Cognito user already onboarded (run `cognito_setup.py` + complete the wizard once; `default-user` account works in `auth-disabled` mode).

Open at: `https://cra-test-alb-1950107331.us-east-1.elb.amazonaws.com/users-admin.html`

X0a. Roles catalogue + SoD reference

1. Open the roles catalogue. Top-right:  Roles catalogue button.

The side drawer surfaces every canonical role with its regulatory basis (ICH E6 §4.1, 21 CFR §312.60–62, Part 11 §11.10, EU CTR Art. 49). Skim:

- **Principal Investigator** — license + GCP + CV + financial disclosure required.
- **Clinical Research Coordinator** — GCP required.
- **Data Manager** — GCP required.
- **Monitor** — sponsor-side; independent from site staff per §5.18.
- **Auditor** — independent from the team being audited per §5.19.

Close the drawer.

X0b. Invite the trial team with scoped role grants

The SMOKE-T2DM-001 trial needs a 5-person team. We'll invite each with the right scope, watching the SoD matrix block the obvious conflicts.

1. Open the Invite modal. Top-right: `+ Invite user`.

2. Invite the PI.

- Email: `pi@smoke.example`
- Role row 1: `Principal Investigator (PI)` · scope `study` · scope id `SMOKE-T2DM-001`

- Click **Send invitation**.

The status pill reads `Invited pi@smoke.example (cognito=FORCE_CHANGE_PASSWORD)`. The Cognito invitation email lands in the sponsor's mailbox; the `PendingInvitation` row carries the scoped grant so first login auto-assigns `principal_investigator @ study:SMOKE-T2DM-001`.

3. Invite the coordinator.

- Email: `coord@smoke.example`
- Role row 1: `Clinical Research Coordinator (CRC) · scope site · scope id SITE-01`
- Click **Send invitation**.

4. Demonstrate an SoD block: try to invite a person as both DM AND PI on the same study.

- Email: `dm@smoke.example`
- Role row 1: `Data Manager (DM) · scope study · scope id SMOKE-T2DM-001`
- Click **+ Add another role**
- Role row 2: `Principal Investigator (PI) · scope study · scope id SMOKE-T2DM-001`
- Notice the **conflict block** appears inline:

***Data Manager (DM) ↔ Principal Investigator (PI)** on study `SMOKE-T2DM-001` 21 CFR Part 11 §11.10(d) + ICH E6 §1.27 — the Data Manager locks the database and the Principal Investigator signs the casebook; they must be distinct individuals to preserve the audit + accountability separation.*

- Click **Send invitation** → 422 with the conflict surfaced. The matrix is enforced server-side; the client-side block is a UX nudge.

5. Fix it — invite just the DM.

- Delete the PI row (× button) so only `Data Manager @ study:SMOKE-T2DM-001` remains.
- Click **Send invitation** → succeeds.

6. Invite the monitor.

- Email: `monitor@smoke.example`
- Role row 1: `Clinical Research Associate (Monitor) · scope study · scope id SMOKE-T2DM-001`
- Click **Send invitation**.

7. Invite the auditor (demonstrates the global-scope path).

- Email: `auditor@smoke.example`
- Role row 1: `Auditor (independent QA) · scope global · scope id (omit)`
- Click **Send invitation**.

8. (Optional) Legitimate SoD override. Single-site academic studies sometimes need the same person to wear two regulatory hats. The platform allows an explicit override with rationale, NEVER silently:

- Open Invite modal → email `dual@smoke.example`
- Role row 1: `Coordinator @ site:SITE-01`

- Role row 2: Monitor @ study:SMOKE-T2DM-001
- Conflict block appears (ICH E6 §5.18 — sponsor monitor cannot also be site staff)
- Fill the **Override rationale** textarea: Investigator-initiated single-site academic study; institutional approval CAS-2026-014 on file. Reviewed by IRB minutes 2026-05-15.
- Click **Send invitation** → succeeds. The rationale is stored on the RoleAssignment row + replicated to the audit log as role.grant_overridden.

X0c. Invitee onboarding — PI walks the wizard

Switch to an incognito browser window (or a separate browser profile).

1. Open the Cognito invite link from `pi@smoke.example`'s mailbox → set a permanent password → land on /.

2. Try to reach `/collector.html` → IIFE redirects to `/onboarding.html` because `/auth/me.onboarding_required: true`.

3. Walk the stepper. The wizard reads the PI's role assignments + `REQUIRED_FIELDS` and renders only the conditional steps the role needs.

For the PI, the visible steps are:

- **Welcome** — shows the regulatory basis for the PI role (ICH E6 §4.1; 21 CFR §312.60+§312.62; EU CTR Art. 49). Click **Next**.
- **Identity** — title Dr, first Ada, last Lovelace, credentials MD MRCP. Save → Next.
- **Licensure** — license number GMC-12345678, country United Kingdom. Save → Next.
- **Training** — completed date 2026-01-15, provider CITI, certificate URL `https://drive.example/cita-cert.pdf`. Save → Next.

Behind the scenes: the wizard POSTs `/api/onboarding/me/training-records` with `training_type=ich_gcp`; the server creates the TrainingRecord row AND mirrors `gcp_training_completed_date + gcp_training_provider + gcp_certificate_url` into UserProfile so the gate sees the GCP field as filled.

- **Disclosure** — CV URL `https://drive.example/cv-pi.pdf`, financial-disclosure signed date 2026-02-01. Save → Next.
- **Delegation** — for each trial-scoped grant, capture delegation entries: - Trial id: (paste the trial id from `/accounts.html` trial detail page) - Study role: Principal Investigator - Delegated tasks: final medical review, AE classification, casebook signature, SAE causality assessment - Start date: 2026-06-01 - Click **Add entry**

The entry is captured *unsigned* — `signed_by_pi_user_id` stays null. The PI countersigns later (X0e).

- **Complete** — review screen. If any field is still missing, the wizard shows AMBER "still missing" chips with the role + field name; the Complete button stays enabled but a 422 from the server kicks the user back. Click **Complete onboarding**.

The server re-validates EVERY required field (anti-tamper), sets `onboarding_completed_at = now()`, records the `onboarding.completed` audit event, and redirects to /.

4. Verify the gate dropped. Click the sidebar + Data Capture (or visit `/collector.html`) → loads. The IIFE no longer redirects because `onboarding_required` is now false.

X0d. Coordinator walks a shorter wizard

The wizard's `activeSteps()` filters conditional steps per role. A coordinator's `REQUIRED_FIELDS = first_name + last_name + gcp_training_completed_date`, so:

1. **Log in as `coord@smoke.example`** (different incognito window).
2. **Steps shown:** Welcome → Identity → Training → Delegation → Complete. (Licensure + Disclosure hidden — not required for coord per ICH E6 §4.1.5.)
3. **Fill Identity + Training** with mock values. Skip Disclosure entirely. Capture a delegation entry on the trial: `study_role Clinical Research Coordinator, tasks informed consent, enrolment, data entry, query response`.
4. **Click Complete** → server-side check passes → gate drops.

X0e. PI countersigns the team's delegation entries

Switch back to the admin window. The PI account (`pi@smoke.example`) now needs to sign the coordinator's + DM's + monitor's delegation entries — ICH E6 §4.1.5 requires the PI signature on the delegation log.

1. **Open `/users-admin.html`** as the PI (PI also holds admin if you grant `user.manage`; otherwise, the PI sign queue is just for visibility). Click the **PI sign queue** tab.
2. **The queue lists every unsigned delegation entry on trials the caller is PI on.** Each row carries the member email + study role + delegated tasks + period + a **Sign** button.
3. **Click Sign** on the coordinator's entry → password modal opens.
4. **Enter the PI's Cognito password** → Submit → 21 CFR Part 11 §11.200 reauth fires against Cognito (the same path eCRF E7 form-signing uses) → on success, the entry is countersigned with `signed_by_pi_user_id + signed_at + the audit event delegation.signed`.
5. **Repeat for each unsigned entry.**

X0f. Audit log inspection

Still on `/users-admin.html`, click the **Audit log** tab.

1. **Click Apply** (no filters). The 6-column table renders the last N events newest-first:

2026-06-01 14:22	pi@smoke	delegation.signed	coord@smoke	trial:SMOKE-...	entry_id=del-..., study_role=Clinical Research Coordinator
2026-06-01 14:15	coord@smoke	onboarding.completed	coord@smoke	—	roles=["coordinator"]
2026-06-01 14:14	coord@smoke	training.recorded	coord@smoke	—	training_type=ich_gcp, topic=ICH E6 GCP
2026-06-01 14:01	admin@smoke	invite.created	—	—	email=coord@smoke. example, assignments=[{role: coordinator, scope_type: site, scope_id: SITE-01}]
2026-06-01 13:58	admin@smoke	role.grant_overridden	dual@smoke	study:SMOKE-...	override_rationale=Investigator-initiated...

2. **Filter for `role.grant\_overridden`** (Action dropdown). Every legitimate SoD override surfaces here — auditors filter on this to inspect every exception.
3. **Click Download PDF** → `user-admin-audit.pdf` lands per Part 11 §11.10(e). Compact-encoded payloads in monospace; truncated at 200 chars per cell so a single long rationale doesn't blow up the layout.

X0g. Regulatory PDF reports

Click the **Reports** tab.

1. **Per-trial delegation log.** Paste the trial id → Download. ICH E6 §4.1.5 format: signed vs unsigned counts at the top, table of (member, role, tasks, period, PI signature). UNSIGNED rows highlight AMBER with a footer

reminder.

2. Per-site training matrix. Paste a site id (`/accounts.html` site picker has the id) → Download. ICH E6 §4.2.4 format: members + expired + expiring-soon counts, table with EXPIRED / DUE SOON AMBER pills on the status cell.

3. Audit log. Click Download full audit (no filters) → same PDF as X0f step 3.

X0h. Suspension demonstration

A user fired / on leave / under investigation needs to be fully blocked from every gated endpoint. The U4 suspension gate (in `web/authz.py.require_permission_scoped`) handles this.

1. Switch to the Users tab on `/users-admin.html`. Locate `coord@smoke.example` → View.

2. Click Suspend. Reason: `Departure from institution; access pending review.` → Submit.

3. Switch to the coordinator's window. Try to load `/collector.html` → 403 with `Your account is suspended.` Contact a platform administrator to reactivate access.

Try to POST data via API: `POST /api/edc/subjects/{id}/forms` → 403 same message.

4. Back in admin → **Click Reactivate** on the coordinator's detail panel → access restored on next request.

X0i. Training expiry surface

Click the **Training expiring** tab.

1. Default within_days = 30. Records with `expires_date` in the past or ≤ 30 days out surface here. The auditor / regulatory-affairs lead uses this to head off lapses before they happen.

2. Click Apply → table renders per-user training records with EXPIRED (red) vs DUE SOON (amber) pills + the topic + provider + completed/expires dates.

Sanity checks

- **SoD enforcement is server-side.** The client-side warning is a UX nudge; the actual 422 fires from `web/user_admin.py.invite` and `grant` even if the operator hits the API directly.
- **Override rationale is REQUIRED to bypass a conflict** and is stored on the RoleAssignment row + replicated to the audit log. NEVER silently accepted.
- **Onboarding gate fires for `Permission` in `services/user\_admin.CLINICAL\_WRITE\_PERMS`** (37 perms: `data.enter`, `query.`, `sdv.verify`, `form.sign`, `casebook.signoff`, `study.lock`, `ae.record/classify`, `screening.`, `visit.update`, `ip.dispense`, `lab.upload`, `source_document.upload`, `cdisc.derive`, `randomization.*`, etc.). Read perms + research-tier skills + admin perms remain open.
- **Suspension gate fires for EVERY perm**, not just clinical-write. It runs BEFORE the onboarding check so the 403 message ("suspended") wins over "onboarding required".
- **PI countersign authority** walks `global` → `trial:<entry.trial_id>` → `study:<id>` (via `EcrfStudy.trial_id`) on the caller's RoleAssignments. A global PI grant sees every unsigned entry; trial/study-scoped PIs see only their scope.
- **Password reauth is Part 11 §11.200.** Hits the same Cognito path the eCRF E7 form-sign endpoint uses. Falls open in auth-disabled tests; runs the real Cognito call in production.
- **The audit log is append-only by convention.** No `update_audit` / `delete_audit` methods on the repo; static-file test pins this.
- **`onboarding.completed` is recorded for every user including admins** — the audit log shows when each member joined the platform's regulatory-grade posture.

- **GCP cert + CV are URL strings**, not file blobs. S3 multipart upload deferred to a polish slice.
- **The wizard re-validates server-side on `POST /me/complete`**. A misbehaving client can't lie about field completeness via DOM manipulation.

X1 — eCRF design (AI-draft CRFs from a protocol) (≈ 5 min)

Demos. AI-drafted CRFs from a protocol paste — designer reviews and edits, never auto-publish.

Open at: <https://cra-test-alb-1950107331.us-east-1.elb.amazonaws.com/ecrf.html>

Steps

1. Click "New study" and fill in:

- **Protocol ID:** SMOKE-T2DM-001
- **Title:** Phase 2 SGLT2i in adults with T2DM (smoke study)
- Click **Create study**.

2. Click "Draft forms from protocol" and paste:

Phase 2, single-arm, open-label study of an SGLT2 inhibitor in adults with type 2 diabetes mellitus.

Eligibility:

- Adults aged 18–75 years.
- HbA1c 7.0–10.0% at screening.
- On stable metformin ≥3 months prior to enrolment.
- BMI 22–40 kg/m².

Schedule of assessments:

- Visit 1 (Baseline, Week 0): consent, demographics, vital signs, fasting glucose, HbA1c, body weight, eligibility verification, study-drug dispensing.
- Visit 2 (Week 4): vital signs, fasting glucose, AE review.
- Visit 3 (Week 12): vital signs, fasting glucose, HbA1c, body weight, AE review, lab safety panel.
- Visit 4 (Week 24, primary endpoint): vital signs, fasting glucose, HbA1c, body weight, AE review, lab safety panel.
- Visit 5 (EOS, Week 28): final vital signs, AE review, treatment-emergent serious AE inventory.

Primary outcome: change in HbA1c from baseline to Week 24.

Secondary outcomes: fasting glucose, body weight, proportion with HbA1c <7.0% at Week 24.

Safety outcomes: vital signs, lab safety panel, treatment-emergent adverse events (graded per CTCAE v5).

Patient-reported outcomes (ePRO, weekly):

- Diabetes Treatment Satisfaction Questionnaire (DTSQ, 8 items).
- Self-reported hypoglycaemia events (frequency + severity).

Click **Draft forms** → StudyDraft returned with proposed forms (Demographics, Eligibility, Vital Signs, Lab Safety, HbA1c, AE Reporting, DTSQ ePRO, etc.) + Visit Schedule. **Nothing auto-saves.**

3. Click "Accept all" (or pick a subset) → forms appear at status=draft.

4. Review + edit a form. Click **Vital Signs**:

- Add a field (+ Item) for systolic BP with edit-check range: 70–250 mmHg, severity: hard.
- Soft check: BP 200 if rule is "warn over 180" → auto-raises a query on save.
- Click **Save draft**.

5. Publish (immutable). Click **Publish** → form moves draft → published. Definition is now immutable; further edits require **New version**.

6. Export ODM-XML. GET `/api/ecrf/forms/{form_id}/export.odm.xml` → CDISC ODM-XML document.

Sanity checks

- **Nothing auto-publishes.** Drafted forms land at `status=draft`.
- **CDASH naming.** Demographics fields like `BRTHTDC` / `SEX` / `RACE` follow CDASH conventions.
- **Item-ID stability.** Renaming a label does NOT change the `item_id`.
- **Hard vs soft checks distinct.** Hard returns 422; soft saves but creates a query.
- **Published forms cannot be edited in place.** PUT `/api/ecrf/forms/{form_id}` on a published form → 409. Must `new-version` first.

X2 — EDC capture (site coordinator) (≈ 5 min)

Pre-req. X1 (published Vital Signs form).

Open at: `https://cra-test-alb-1950107331.us-east-1.elb.amazonaws.com/collector.html`

Steps

1. Create a deployment. Click **New deployment** → select the published study → name it `Smoke deployment`. This snapshots published form definitions into `DeployedForm` rows.

2. Add a site: `Site 01 – University Hospital`.

3. Add a subject: `subject_code S01-001`.

4. Open a form for the subject: click the subject's row → pick `Vital Signs (Visit 1, Baseline)` → creates a `FormInstance` at `status=in_progress`.

5. Enter data → save → observe edit-checks:

- Type values. Violate a hard check (BP 300) → 422 with structured failure list `{detail: {failures: [{item_id, check_id, message}]}}`. Nothing persisted.
- Fix → save succeeds; persists in `ItemData` rows.
- Violate a soft check (BP 200) → saves but **auto-raises a `Query`** in the form's query panel.

6. Query workflow: Open the **Queries** panel → respond / close queries. Manual queries: POST `/api/edc/form-instances/{form_instance_id}/queries`.

Sanity checks

- **Hard check = blocked save**, 422 response, no `ItemData` written.
- **Soft check = auto-query**, referencing offending `item_id` + check that fired.
- **Audit trail accumulates.** Every save records `AuditEntry` with `old_value` → `new_value`, `actor_sub`, `source=edc`. Inspect: GET `/api/edc/form-instances/{form_instance_id}/audit`.

X3 — ePRO capture (participant magic-link) (≈ 5 min)

Pre-req. X1 (published DTSQ form flagged `epro=true`) + X2 (subject S01-001).

Steps

1. Issue a magic link. From the subject's row in `collector.html` → **Issue ePRO link** → POST `/api/edc/subjects/{subject_id}/epro-access` returns `{token, epro_path: "/epro.html?token=..."}`. Raw token shown **once**; hashed in DB.

2. Open the participant surface in an incognito window:

`https://cra-test-alb-1950107331.us-east-1.elb.amazonaws.com/epro.html?token=<the-issued-token>`

Consent gate appears first → tap **I consent** → POST `/api/epro/consent?token=...` records timestamp.

3. Pick a PRO form (DTSQ weekly) → enter 8 items → **Submit**. Set `mark_complete=true` on save to lock the instance.

Sanity checks

- **Token scope is per-subject.** Trying to open a form for a different subject → 403.
- **No coordinator data exposed.** ePRO surface only renders forms flagged `epro=true`. Site-only forms invisible.
- **Consent permanent.** Recorded timestamp persists.
- **No raw token in DB.** Only the hash.

X4 — AE / SAE workflow + auto-classification + FDA 3500A (≈ 10 min)

Demos. ICH E2A auto-classification on every AE write, 24h escalation timer, FDA 3500A IND safety report draft.

Pre-req. SMOKE deployment + subject S01-001 from X1-X2.

Steps

1. Capture an AE (coordinator). In `collector.html` → Safety panel → + **Adverse Event**:

```
{
  "subject_id": "<S01-001 id>",
  "term_text": "Severe hypoglycaemic episode requiring ED visit",
  "meddra_pt": "Hypoglycaemia",
  "start_date": "2026-05-31T08:30:00Z",
  "severity_grade": 4,
  "outcome": "recovering",
  "relationship_to_intervention": "probable"
}
```

→ on save, `safety_rules.auto_classify_serious(...)` fires:

- `is_serious=True` (severity 4 = life-threatening)
- `serious_reasons=["life_threatening", "hospitalisation"]`
- `reportable_deadline = now() + 24h`

2. PI review (PI account). PATCH the AE to confirm classification:

```
PATCH /api/edc/ae/{ae_id}
{
  "is_serious": true,
  "pi_classified_at": "2026-05-31T09:00:00Z"
}
```

3. Check overdue-SAE dashboard. GET `/api/edc/ae/overdue` → returns any AEs past their `reportable_deadline` without `reported_to_authority_at` set.

4. Generate FDA 3500A draft. Click **Draft FDA 3500A** → `reports/sae_3500a.py` renders PDF with platform-derived fields filled + sponsor-supplied fields marked [SPONSOR INPUT REQUIRED].

5. PI downgrades severity (optional). PATCH severity to 2 → auto-clears `reportable_deadline`; audit row records the downgrade.

Sanity checks

- ``ae.record`` is **coordinator-tier**, `ae.classify` is PI-tier. Coordinator PATCH attempt → 403.
- ``is_serious`` + ``serious_reasons`` **persisted at write time**, NOT recomputed on read. A rule change after the fact doesn't quietly re-classify history.
- ``reportable_deadline`` **set to now()+24h** on first serious classification. Cleared on downgrade.
- **3500A masks PHI** — only `subject_code`, never participant name.
- **MedDRA PT free-text MVP.** Real validation requires MedDRA license at deploy time.

X5 — Protocol deviation + CAPA lifecycle (≈ 5 min)

Demos. Major / minor / critical classification + CAPA author + close.

Steps

1. Log a deviation (coordinator).

```
POST /api/edc/deployments/{deployment_id}/deviations
{
  "subject_id": "<S01-001 id>",
  "category": "visit_window",
  "description": "Visit 3 occurred 11 days outside window due to subject travel"
}
```

2. Classify (DM).

```
PATCH /api/edc/deviations/{deviation_id}/classify
{ "severity": "minor", "rationale": "Outside window but no impact on endpoint" }
```

3. Author a CAPA (DM).

```
POST /api/edc/deviations/{deviation_id}/capas
{ "action_text": "Re-train site coordinator on visit-window definitions", "owner_sub": "<DM sub>", "due_date": "2026-06-15" }
```

4. Complete CAPA + close deviation (PI).

```
POST /api/edc/capas/{capa_id}/complete
POST /api/edc/deviations/{deviation_id}/close
```

Sanity checks

- ``deviation.classify`` is DM; ``capa.close`` is PI.
- **PI cannot close deviation while any CAPA is `open`** → 422.
- Audit trail per state transition.

X6 — Recruitment / screening log + CONSORT funnel (≈ 8 min)

Demos. Screening → eligible → consented → enrolled funnel + per-reason exclusion histogram.

Open at: `collector.html` → Recruitment panel.

Steps

1. Record screenings (coordinator). For each prospective subject:

```
POST /api/edc/deployments/{deployment_id}/screening-logs
{
  "screening_code": "SCR-0001",
  "site_id": "<Site 01 id>",
  "age_band": "40-49",
  "sex": "F",
  "race": "white"
}
```

Repeat with varying outcomes — for the exclusion path:

```
PATCH /api/edc/screening-logs/{log_id}/eligibility
{ "eligibility_status": "screen_failure", "exclusion_reason_code": "hba1c_out_of_range" }
```

For the consent + enrolment path:

```
PATCH /api/edc/screening-logs/{log_id}/eligibility { "eligibility_status": "eligible" }
PATCH /api/edc/screening-logs/{log_id}/consent { "consent_status": "consented", "consent_date": "2026-05-31" }
PATCH /api/edc/screening-logs/{log_id}/enrolment { "enrolment_status": "enrolled", "enrolled_subject_id": "<S01-001 id>" }
```

2. View the funnel rollup. GET /api/edc/deployments/{deployment_id}/recruitment-funnel returns:

- 4 stage counts (screened / eligible / consented / enrolled).
- Per-reason exclusion histogram (8 CONSORT-aligned codes + 'other').
- Per-week × per-site stage counts.

3. Collector.html Recruitment panel surfaces the funnel + recent screenings + a record form.

Sanity checks

- **State machines monotonic.** Cannot consent before `eligible`, cannot enrol before `consented`. Repo raises `ClinicalError`.
- **8 canonical exclusion reasons** (`consent_refused` / `age_out_of_range` / `disease_severity` / `contraindication` / `not_meeting_disease_criteria` / `unable_to_comply` / `lost_to_contact` / `other`).
- **PHI minimisation.** `age_band` + `sex` + `race` + `dob_year` only — no full DOB / MRN.
- **Funnel rollup computed in Python** for SQLite/Postgres portability.

X7 — Visit scheduling + reminders (≈ 8 min)

Demos. VisitSchedule → ScheduledVisit → PlannedVisit auto-gen at enrolment → email reminders.

Steps

1. Create a visit schedule (study_designer).

```
POST /api/edc/deployments/{deployment_id}/visit-schedules
{ "name": "Main protocol", "description": "5-visit schedule per protocol" }
```

2. Add scheduled visits.

```
POST /api/edc/visit-schedules/{schedule_id}/visits
{ "visit_name": "Baseline (Visit 1)", "day_offset": 0, "window_before_days": 3, "window_after_days": 3, "reminder_offsets_days": [-3, -1] }
```

```
POST /api/edc/visit-schedules/{schedule_id}/visits
{ "visit_name": "Week 4 (Visit 2)", "day_offset": 28, "window_before_days": 5, "window_after_days": 5,
  "reminder_offsets_days": [-7, -3, -1] }
```

(Repeat for Visits 3, 4, 5 at day_offsets 84, 168, 196.)

3. Activate the schedule (DM):

```
POST /api/edc/visit-schedules/{schedule_id}/activate
```

4. Set subject baseline date + generate planned visits:

```
PATCH /api/edc/subjects/{subject_id}/baseline-date { "baseline_date": "2026-06-01T00:00:00Z" }
POST /api/edc/subjects/{subject_id}/planned-visits/generate
```

5 PlannedVisits land for S01-001 at the schedule-implied dates.

5. Register a participant contact:

```
POST /api/edc/subjects/{subject_id}/participant-contact
{ "email": "participant@example.com", "opt_in_channels": ["email"] }
```

6. Trigger reminders manually:

```
POST /api/edc/deployments/{deployment_id}/reminders/fire-due
```

→ returns {queued: N, sent: M, failed: K, skipped: 0}. If `AWS_SES_FROM_EMAIL` not set, `provider='dry_run'` is logged.

Sanity checks

- **Reminder offsets must be non-positive ints** (days BEFORE due_date). Positive → 422.
- **Planned-visit generation idempotent** per subject × scheduled_visit (UNIQUE).
- **PlannedVisit window** = planned_date ± window_before/after_days.
- **SES dry-run mode** writes `SentReminder` with `status='sent'`, `provider='dry_run'` when env var unset.
- **SMS reminders** logged with `status='skipped'` (Twilio not wired).
- **Reschedule** requires non-empty `override_reason`.

X8 — Drug accountability (≈ 8 min)

Demos. Per-lot inventory + dispense/return state invariants + reconciliation rollup.

Steps

1. Register an investigational product (study_designer or DM):

```
POST /api/edc/deployments/{deployment_id}/ip-catalogue
{ "drug_name": "Empagliflozin", "strength": "10 mg", "units": "tablet" }
```

2. Receive a shipment (coordinator or DM):

```
POST /api/edc/deployments/{deployment_id}/drug-receipts
{ "ip_id": "<ip_id>", "lot_number": "LOT-A1", "quantity_received": 1000, "site_id": "<Site 01 id>" }
```

3. Dispense to subject (coordinator or PI):

```
POST /api/edc/deployments/{deployment_id}/drug-dispensations
{ "subject_id": "<S01-001 id>", "ip_id": "<ip_id>", "lot_number": "LOT-A1", "kit_id": "KIT-001",
  "quantity_dispensed": 30 }
```

4. Try to over-dispense:

```
POST /api/edc/deployments/{deployment_id}/drug-dispensations
{ "subject_id": "<S01-001 id>", "ip_id": "<ip_id>", "lot_number": "LOT-A1", "kit_id": "KIT-002",
"quantity_dispensed": 5000 }
```

→ 422: "Cannot dispense 5000 — only 970 tablet(s) of lot LOT-A1 in inventory."

5. Log a return (coordinator):

```
POST /api/edc/drug-dispensations/{dispensation_id}/return
{ "quantity_returned": 10, "quantity_used": 18, "quantity_lost": 0, "return_reason": "end_of_visit" }
```

→ 422: "quantity_used + quantity_lost cannot exceed quantity_returned." Adjust:

```
{ "quantity_returned": 18, "quantity_used": 12, "quantity_lost": 0, "return_reason": "end_of_visit" }
```

6. View reconciliation rollup (DM / monitor / auditor / PI):

```
GET /api/edc/deployments/{deployment_id}/drug-reconciliation
```

→ per-lot inventory: `current_inventory = received + (returned - used - lost) - dispensed`.

Sanity checks

- **Dispense over inventory** → 422. Inventory derived (not stored).
- **Return without prior dispensation** → 422 (FK on dispensation_id).
- **quantity_used + quantity_lost > quantity_returned** → 422.
- **Cross-deployment subject vs IP** → 422.
- **`ip.reconcile` gates the rollup endpoint.** Coordinator GET → 403.

X9 — Source-document extraction (CSV → eCRF) (≈ 10 min)

Demos. EHR / registry CSV → versioned mapping → audit-traced ExtractionFill.

Steps

1. Upload a source CSV (coordinator). Prepare a minimal CSV:

```
patient_id,visit_date,sbp_mmhg,dbp_mmhg,heart_rate_bpm,weight_kg
S01-001,2026-06-01,128,82,72,84.5
S01-001,2026-06-29,125,80,70,84.0
```

Upload via `collector.html` Source-extraction panel:

```
POST /api/edc/deployments/{deployment_id}/source-documents (multipart)
```

→ `SourceDocument` row with SHA-256 content hash + N parsed rows.

2. Author an extraction mapping (study_designer or DM).

```
POST /api/edc/deployments/{deployment_id}/extraction-mappings
{
  "deployed_form_id": "<Vital Signs form id>",
  "name": "default",
  "subject_code_field": "patient_id",
  "mapping": {
    "sbp_mmhg": "sbp",
    "dbp_mmhg": "dbp",
    "heart_rate_bpm": "hr",
    "weight_kg": "weight"
  }
}
```

3. Dry-run the mapping (coordinator or DM).

POST /api/edc/extraction-mappings/{mapping_id}/dry-run?source_document_id={doc_id}

→ returns the rows that would be written, no persistence yet.

4. Apply the mapping in `subjects` mode (pre-fills eCRF instances).

POST /api/edc/extraction-mappings/{mapping_id}/apply
{ "source_document_id": "<doc_id>", "mode": "subjects" }

→ {subjects_filled: 1, items_written: 8, source_rows_unmatched: 0}. Existing FormInstances for S01-001 receive ItemData with audit chain.

5. Inspect per-cell audit (DM / monitor / auditor):

GET /api/edc/deployments/{deployment_id}/extraction-fills?target_id=<subject_id>

→ each ExtractionFill row carries source_row_id, source_field, mapping_version (denormalised so audit survives mapping mutations), applied_value, applied_by_sub, applied_at.

6. Re-upload the same file → SHA-256 dedupes; returns existing SourceDocument row.

Sanity checks

- **SHA-256 dedupes.** Same content → existing doc returned.
- **Versioned mappings.** Authoring a new mapping for the same (deployment, form) auto-deactivates the prior version.
- **`mapping\_version` denormalised** in ExtractionFill so audit chain survives mapping mutations.
- **RESTRICT FK on mapping** prevents deletion while audit history exists.
- **Dual modes.** mode="subjects" pre-fills eCRF; mode="table" returns a flat projection without persistence.

X10 — Lab-data feeds (HL7 v2 + CDISC LAB + FHIR) (≈ 10 min)

Demos. Three stdlib parsers, idempotent SHA-256 ingest, SDTM LB cascade.

X10a. HL7 v2 ORU^R01 upload

1. Save the following to `cbc.hl7`:

```
MSH|^~\&|LAB|HOSP|EDC|TRIAL|20260531120000||ORU^R01|MSG1|P|2.5  
PID|||S01-001  
OBR|1||ACC123|CBC^Complete Blood Count^L|||20260531110000  
OBX|1|NM|HGB^Hemoglobin^L||14.2|g/dL|13.0-17.0|N|||F  
OBX|2|NM|GLUC^Glucose^L||102|mg/dL|70-100|H|||F
```

2. Upload via collector.html Lab-data feeds panel:

POST /api/edc/deployments/{deployment_id}/lab-uploads (multipart, source_format=hl7v2)

→ LabBatch.row_count=2 + was_new=True.

3. Verify subject linkage. GET /api/edc/lab-batches/{batch_id}/results → both rows have subject_id set (S01-001 was registered before ingest).

X10b. Re-upload (dedupe)

Re-POST the same file → returns existing batch with was_new=False. Confirm only 2 rows total in GET /api/edc/deployments/{id}/lab-results.

X10c. CDISC LAB tab-delimited

Save as lab.tsv:

STUDYID■SITEID■SUBJID■ACCSNUM■LBTESTCD■LBTEST■LBORRES■LBORRESU■LBORNRL0■LBORNRI■LBNRIND■LBOTC
 STUDY1■01■S01-001■A1■CREAT■Creatinine■0.9■mg/dL■0.7■1.3■NORMAL■2026-05-31T11:05:00
 STUDY1■01■S01-001■A1■BUN■Urea Nitrogen■18■mg/dL■7■20■NORMAL■2026-05-31T11:05:00

Upload with `source_format=cdisc_lab`.

X10d. FHIR R4 Bundle (listener endpoint)

Save as `obs.json`:

```
{
  "resourceType": "Bundle",
  "type": "collection",
  "entry": [
    {
      "resource": {
        "resourceType": "Observation",
        "subject": { "reference": "Patient/S01-001" },
        "code": { "coding": [{ "code": "TSH", "display": "TSH" }] },
        "valueQuantity": { "value": 2.1, "unit": "mIU/L" },
        "referenceRange": [{ "low": { "value": 0.4 }, "high": { "value": 4.0 } }],
        "effectiveDateTime": "2026-05-31T11:05:00Z"
      }
    }
  ]
}
```

POST to the listener endpoint with the right Content-Type:

POST /api/edc/deployments/{deployment_id}/lab-feed
 Content-Type: application/fhir+json
 <the JSON body>

X10e. SDTM LB cascade

Re-derive CDISC:

POST /api/edc/deployments/{deployment_id}/cdisc/derive

→ Download `sdm/lb.csv` from the Submissions card → confirm LB rows include the HL7 + CDISC LAB + FHIR results (USUBJID=<study_id>-S01-001, LBSEQ unique per subject, LBNRIND derived).

Sanity checks

- **SHA-256 dedupe** by (`deployment_id`, `content_hash`).
- **Subject linkage.** `subject_code_hint` preserved when no Subject match yet; POST `.../lab-results/backfill-subjects` re-links after a Subject is created.
- **Empty payload** → 422. Unknown format → 422.
- **Listener Content-Type** → **source_format mapping.** `application/hl7-v2+er7` → `hl7v2`, `application/fhir+json` → `fhir`, `text/tab-separated-values` → `cdisc_lab`.
- **SDTM LB cascade additive.** Form-based `derive_lb` runs first; parsed labs append with `starting_seq_by_subject` continuation.

X11 — Multi-site rollout dashboard (≈ 5 min)

Demos. Per-site KPIs across enrolment / queries / safety / operations.

Pre-req. A deployment with ≥2 sites + some screenings + queries + AEs + planned visits + lab batches.

Open at: <https://cra-test-alb-1950107331.us-east-1.elb.amazonaws.com/multisite.html>

Steps

1. Pick the **SMOKE** deployment from the deployment picker.

2. Each site card surfaces 4 KPI groups:

- **Enrolment funnel** — screened / eligible / consented / enrolled.
- **Query backlog** — open / answered / closed.
- **Safety** — open AEs / open serious AEs / open deviations / open CAPAs.
- **Operational** — overdue visits / low-IP lots / last SDV timestamp.

3. Tune the **low-IP threshold** via the input field (default 10).

4. **Admin-only org rollup.** Switch to an admin account → an **Organisation rollup** section appears at the top with one card per deployment + org-wide totals. Click a deployment card → drills into its per-site view.

Sanity checks

- **Pure read-side aggregation** — no new tables, no new columns.
- **Open-AE definition.** `outcome NOT IN ('recovered', 'death')`. Recovering / not_recovered / unknown / default all count as open.
- **Overdue visit** = ``status='pending' AND window_end < now``.
- **Org rollup gated on** ``portfolio.read_org`` (admin only).
- **Per-deployment rollup gated on** ``study.read`` at deployment scope.
- ``low_ip_lots`` surface on totals row, NOT per-site cards (IP is deployment-scoped today).

X12 — Sign / lock / SDV / audit (≈ 5 min, partly API-only)

Demos. Form-instance signing → casebook sign-off → SDV → unlock voids signature with audit.

Pre-req. Completed FormInstance from X2.

```
POST /api/edc/form-instances/{form_instance_id}/sign {"meaning": "author", "password": "<re-auth>"}
POST /api/edc/form-instances/{form_instance_id}/verify (monitor, source-data verification)
POST /api/edc/subjects/{subject_id}/sign {"meaning": "casebook"}
POST /api/edc/form-instances/{form_instance_id}/unlock {"reason": "correction"}
GET /api/edc/form-instances/{form_instance_id}/signatures
GET /api/edc/form-instances/{form_instance_id}/audit
```

Sanity checks

- **Signing requires password re-auth** (Part 11 §11.200) — verified against Cognito.
- **Signing locks the form.** PUT new ItemData → 409. Signature bound to data hash at sign time.
- **Unlock voids the signature** (`voided=True`) but keeps the row + writes a new AuditEntry.
- **Append-only audit.** No UPDATE or DELETE statements ever touch AuditEntry. DB-level constraint enforces this (E6). Try `UPDATE audit_entries ...` in psql → rejected.
- **PHI segregation.** Clinical tables — Subject + FormInstance + ItemData + Signature + Query + AuditEntry — live **only** in the separate clinical database, never in the research-app database.

X13 — eCRF validation pack (IQ / OQ / PQ + study lock) (≈ 8 min)

Demos. "Deployable in an audited environment" gate — IQ snapshot + OQ pytest run + PQ runbook + ZIP bundle + study-level database lock.

Steps

1. Lock the database (DM).

```
POST /api/edc/deployments/{deployment_id}/lock
{ "reason": "Pre-CSR database lock" }
```

→ tries to lock. If any non-closed queries exist, fails with 422 listing the queries. Override with `force_open_queries: true` (audited):

```
POST /api/edc/deployments/{deployment_id}/lock
{ "reason": "Pre-CSR database lock", "force_open_queries": true }
```

2. Try to write data while locked. Attempt any PUT / POST against `/api/edc/.../data` → 409 with "Study deployment X is locked".

3. View lock status + history:

```
GET /api/edc/deployments/{deployment_id}/lock-status
GET /api/edc/deployments/{deployment_id}/lock-history
```

4. Download the validation pack (admin). From `admin.html` → Validation Pack section:

- **IQ.pdf** — package version + Python version + pinned dependencies from `uv.lock` + Cognito pool id + audit-trigger detection.
- **OQ.pdf** — shells out to pytest over the test node-ids declared in the Requirements Traceability Matrix (`validation/requirements_matrix.json`) and joins pass/fail with each requirement's regulatory anchor (Part 11 §, ICH E6, ICH E2A, ALCOA+).
- **PQ.pdf** — 8-step customer-side runbook (login → study create → form sign with reauth → study lock → IQ download → OQ run).
- **bundle.zip** — all three PDFs + the RTM JSON.

5. Unlock (DM only):

```
POST /api/edc/deployments/{deployment_id}/unlock
{ "reason": "Found a missing AE – needs to be entered before CSR" }
```

Sanity checks

- **`study.lock` is DM-only.** PI attempt → 403 (separation of duties: PI signs casebook, DM locks DB).
- **Locked deployment refuses every mutation** — PUT data, POST sign, POST verify, POST AE record, POST deviation record, POST query raise, etc. all 409.
- **CDISC derivation is allowed post-lock** (lock-then-derive is regulator-default).
- **OQ requirement traceability.** Each row in the OQ PDF carries Part 11 § / ICH E6 / ICH E2A / ALCOA+ anchor.

Phase 05 — Analysis & reporting

***Phase context.** After database lock, the team runs the planned statistical analyses, assembles the CDISC submission bundle, writes the CSR, drafts the manuscript, and produces patient-facing materials.*

A1 — Trial-specific statistical analysis (KM + Cox + MMRM + binary + subgroup) (≈ 10 min)

Demos. 7-stage post-lock workflow, sandbox-backed, every result row carries a `derived_from` audit anchor.

Trigger:

`/trial-stats run efficacy + safety analyses on the SGLT2i SMOKE-T2DM-001 ADaM bundle, primary endpoint HbA1c at Week 24`

Expected flow: 7 stages — intake → analysis populations (ITT / mITT / PP / Safety) → time-to-event (Cox PH + K-M + log-rank) → continuous (MMRM with TRT×visit) → binary (Fisher + log-binomial GLM) → subgroup (per-subgroup Cox + interaction p + forest PNG) → assembled document.

Operator pastes ADSL + ADTTE + ADLB extracts at each stage; the model calls `run_trial_analysis(kind, payload)` which fans out to `kaplan_meier.py` / `mmrm.py` / `binary.py` / `subgroup_forest.py` in the sandbox.

Sanity checks

- Every result row carries `derived_from = "sandbox:<kind>:<paramcd>"`. HRs / LSMean diffs / CIs / p-values cannot be inline-asserted.
- MMRM fits via `statsmodels.MixedLM` with visit fixed effects + TRT × visit + optional BASE covariate.
- Cox PH fits via `statsmodels.duration.PHReg`.
- Subgroup forest PNG embedded in PDF; per-subgroup HR + interaction p surfaced.
- `TrialStatsDocument.csr_artefact_ids`` is a `@property` the CSR drafter cites.

Cross-handoff

Click → **Draft CSR from trial-stats** on the assembled card → seeds the CSR drafter (see A4) with the trial-stats document JSON.

A2 — Beyond-forest-plot visualisations (funnel + waterfall + swimmer + GRADE chip) (≈ 8 min)

Demos. Four plot types across three specialists; sandbox-rendered diagnostics + host-rendered GRADE chip.

Funnel + Egger's test

Built into meta-analysis STEP 5 (E1 already exposes this). New `funnel.py` script uses `statsmodels.api.OLS` for Egger's regression. Each outcome with ≥3 studies gets `funnel_plot_image` + `eggerts_p_value` + `funnel_interpretation`.

Waterfall (per-subject best response)

Trigger after a trial-stats workflow:

Add waterfall: per-subject HbA1c percent-change from baseline, threshold ±10%

→ `waterfall.py` script renders RECIST 1.1 categorised bars (CR / PR / SD / PD).

`WaterfallResult.derived_from = "sandbox:waterfall:<outcome>"`.

Swimmer (treatment timeline)

Add swimmer: per-subject treatment timeline with `response_onset` / `cr` / progression event markers

→ `swimmer.py` renders horizontal bars with event markers + ongoing-treatment arrows.

GRADE chip table (host-side)

Auto-generated host-side from `GradeAssessment` rows (no agent involvement). Rows = outcomes; columns = 5 downgrade domains + 3 upgrade domains + computed certainty. Each cell colour-coded per the GRADE-pro convention. Embedded in GRADE PDF + DOCX.

Sanity checks

- **No new dependency.** `statsmodels` / `matplotlib` already pinned in sandbox.
- **Chip SVG hand-rolled with ``html.escape``** for XML safety.
- **NMA geometry PNG** lives in the NMA specialist (E4), not here.

A3 — CDISC SDTM → ADaM → TLF submission pipeline (≈ 10 min)

Demos. Full submission bundle — 7 SDTM domains + ADSL + ADTTE + TLF + Define-XML v2.1 + SAS Transport v5.

Pre-req. Smoke deployment with screenings + AEs + lab feeds + at least one `FormInstance` per domain.

Steps

1. Map forms to domains (one-time per deployment, study_designer). Configure `ItemMappingConfig.form_to_domain_map`:

```
{
  "vital_signs": "VS",
  "labs": "LB",
  "exposure": "EX",
  "concomitant_meds": "CM",
  "medical_history": "MH",
  "demographics_form": "DM",
  "ae_form": "AE"
}
```

2. Derive (DM):

`POST /api/edc/deployments/{deployment_id}/cdisc/derive`

→ runs the pipeline: 7 SDTM domains + ADSL + ADTTE. Submission bundle assembled in-process.

3. Render survival TLF (optional):

`POST /api/edc/deployments/{deployment_id}/cdisc/survival/render`

→ sandbox runs `statsmodels` K-M + Cox PH stratified by TRT01A; writes K-M PNGs back as data-URI `TlfArtefact.svg_content` + a Cox PH summary table.

4. Download the bundle. Submissions card in `collector.html` → **Download bundle.zip**:

- `define.xml` at root (CDISC ODM/Define-XML 2.1 covering all 9 datasets with `ItemGroupDefs` + `ItemDefs` + `CodeLists` + `MethodDefs`).
- `define-overview.txt` (human summary).
- `sdtm/*. {csv, xpt}` × 7 (DM / AE / VS / LB / EX / CM / MH).
- `adam/{adsl, adtte}. {csv, xpt}`.
- TLF tables + figures.

Sanity checks

- **Hand-rolled XPT writer** — IBM-360 8-byte float conversion + 80-byte record alignment + 140-byte v5 namestr records. ASCII-only CHAR + NUM; 8-char column-name limit + 200-char string limit enforced.
- **No new dependency.** `xport` / `pyreadstat` deliberately avoided.
- **Define-XML v2.1** validates against the CDISC ODM schema.
- **ADSL.TRT01P / TRT01A** populated from `Allocation.arm` (S3) — no "TBD".
- **ADTTE rows** carry SRCDOM/SRCVAR audit anchors per regulator requirement.
- **MedDRA + WHODrug remain deploy-time gates** — PT / DECOD captured as free-text MVP.

A4 — CSR (ICH E3) drafter (≈ 10 min)

Demos. ICH E3 synopsis + 4 data-driven sections (Disposition / Demographics / Efficacy / Safety) populated from ADSL + ADTTE + TLF. Strictest anti-hallucination posture in the codebase.

Trigger (or cross-handoff from A1 / A3):

`/csr` Phase 2 study of Empagliflozin in adults with T2DM, lock date 2026-05-31, double-blind, FDA submission

Expected flow: 4-stage workflow — intake → synopsis → 4 data sections → assembled document.

The data sections require the operator to paste source-artefact ids (e.g. TLF `t-disposition`, ADTTE `PARAMCD=TTAE`); every count, effect size, CI, p-value must cite its `derived_from` source. Narrative sections (Introduction / Discussion / Overall Conclusions) ship as `[Operator to complete]` placeholders.

Sanity checks

- **Every count / effect size / CI / p-value carries ``derived_from``.** Try to type "HR 0.81 (0.71, 0.93)" inline without a source → validator rejects with "Effect sizes must cite an ADTTE / TLF / TrialStats artefact id."
- **Narrative placeholders honest.** Introduction / Discussion / Overall Conclusions show `[Operator to complete]` colour-coded amber.
- **Skeleton CSR carries all 16 ICH E3 section headers.**
- Continuation prefixes disambiguated (`CSR intake confirmed`, `CSR synopsis confirmed`, etc) so they don't collide with the generic registration prefix.

A5 — Manuscript drafter (IMRaD) + reviewer-response (≈ 8 min)

Demos. Compose a meta-analysis into a journal-ready paper; iterate point-by-point on reviewer comments with a cover letter.

Trigger (or click → `Draft as manuscript` on E1's terminal card):

`/manuscript` compose an IMRaD manuscript from my last meta-analysis, target NEJM

Expected flow: 3-stage workflow — intake → IMRaD draft → reviewer-response.

After Stage 2, paste a fake reviewer comment block:

Reviewer 1:

- The forest plot doesn't include a leave-one-out sensitivity analysis. Please add.
- Some included studies appear to have high risk of bias. How did this affect conclusions?

Reviewer 2:

- The discussion is too long. Please cut by 30%.

The drafter responds with **point-by-point response** + revised manuscript + cover letter. Targets: NEJM / Lancet / BMJ / JAMA / Annals / PLOS ONE / generic.

Sanity checks

- **Journal target enforced** — schema validates against the 7-value enum.
- **Cover letter auto-drafts** with sponsor + word count + financial disclosure placeholders.
- **PDF + DOCX export** bundles cover letter + point-by-point response + revised manuscript.

A6 — Lay summary (3 variants: recruitment + evidence + results) (≈ 10 min)

Demos. Host-side Flesch-Kincaid + compute-and-iterate readability loop + multilingual.

A6a. Recruitment lay summary (from protocol)

/lay-summary plain-language recruitment material at grade 6 English for parents of children with asthma – Phase 2 RCT of inhaled corticosteroid X vs standard care

Intake = a protocol synopsis. Drafts a recruitment-language summary at the configured FK grade.

A6b. Evidence lay summary (from meta-analysis)

Click → **Draft lay summary** on E1's meta-analysis terminal card. The handoff seed carries the meta-analysis JSON. The lay summary cites the underlying PMIDs verbatim.

A6c. Results lay summary (from CSR)

After A4 completes, trigger:

Draft lay summary from CSR

Carries the CSR's primary outcome summary + safety summary + `derived_from` ids forward.

Sanity checks

- **Host-side Flesch-Kincaid grade** computed in `services/readability.py` (stdlib only — no `textstat` dep). The model's self-reported grade is **overwritten** by the host's number.
- **Retry cap = 3.** When first draft scores > 8.0, the specialist re-prompts with simplification directive. `readability_attempts` records 1, 2, or 3.
- **AudienceProfile.** `target_grade` 4-12 (default 6); `language` en/es/fr/de; free-text `region`; `population_descriptor`.
- **5 CISCPR sections** schema-enforced.
- **Anti-hallucination per variant.** Evidence cites only from `pmid_sources`. Results cites only from `derived_from_ids`. Recruitment never promises trial success.
- **No medical-decision language.** "You should" / "you must" forbidden; "may help" / "might give doctors more information" allowed.
- **PDF + DOCX export** with reading-grade badge in cover band + glossary sidebar + "not medical advice" disclaimer footer.

A7 — Citation manager (BibTeX + RIS round-trip) (≈ 5 min)

Demos. Pure file-format round-trip covering Zotero / EndNote / Mendeley.

Steps

1. **Export from Zotero / EndNote / Mendeley** as `.bib` or `.ris`. Or use this minimal `test.bib`:

```
@article{anderson2014,  
  author = {Anderson, J. and Smith, K.},  
  title = {Proton Pump Inhibitor Use in ACS},  
  journal = {Lancet},  
  year = {2014},  
  pmid = {25040175}  
}
```

2. In a manuscript_drafter or sr_protocol intake card, drop the file on the `<CitationImporter>` drop-zone. Or POST directly:

POST /api/citations/parse (multipart file)

→ returns `{citations: [...]}`.

3. **Inline citations** in the next chat message; the specialist uses them as bibliography seed.

4. **Export back.** From the terminal card → **Export bibliography** → choose BibTeX or RIS → downloads file.

Sanity checks

- **Stdlib-only parsers** — no `bibtexparser` / `pyzotero` / `rispy` dependency.
- **5MB upload cap.**
- **Round-trip preservation.** `article` / `book` / `incollection` / `inproceedings` / `techreport` / `phdthesis` / `misc` map across BibTeX / RIS via `_BIBTEX_TYPE_BY_RIS_TY` / `_RIS_TY_BY_BIBTEX_TYPE`.
- **Unknown BibTeX fields preserved** through export via `raw_fields`.
- **PMID detection from RIS** `AN` / `ID` / `PM` tags.

Phase 06 — Cross-cutting infrastructure

Phase context. Behind every phase is the same underlying platform — *auth*, *RBAC*, *library* + *retrieval*, *sandboxing*, *portfolio* + *budget tracking*, and *configuration*.

C1 — Portfolio dashboard (≈ 3 min)

Open at: <https://cra-test-alb-1950107331.us-east-1.elb.amazonaws.com/portfolio.html>

What to demo

- **Hero cards** — total threads / SR projects / deployments / top workflow.
- **Recent threads** table with cost per thread + workflow pill + last activity.
- **SR projects** rollup (owned + `sr_review-membership` combined).
- **eCRF deployments** rollup resolved through `RoleAssignment` scope (global / study / site).
- **Admin org section** auto-appears for admin accounts (gated on `portfolio.read_org`).

Sanity checks

- **Researcher sees own data only.** Non-admin → org section hidden.
- **No row-level data leaked in org rollup** — only counts + roles.

- ``last_turn_kind`` peeks at the most recent assistant `Message.final_answer JSON` for the kind discriminator.

C2 — Budget + cost rollup (≈ 3 min)

Same `portfolio.html` page → "Spend (USD)" section.

What to demo

- **Four hero cards** — This month / Cumulative / Turns / Tokens.
- **By-workflow breakdown** — per-specialist USD spend.
- **By-model-family breakdown** — Haiku / Sonnet / Opus split.
- **Cost column on threads table.**
- **Admin-only Top-spend users** section.

Sanity checks

- **Pricing-table-as-code** in `config/bedrock_pricing.py` — Claude 4.x (Haiku 4.5 / Sonnet 4.6 / Opus 4.7) + legacy 3.x.
- **Pricing provenance documented** in UI ("AWS Bedrock public rates as of 2026-05. Verify before invoicing.").
- **Six-decimal rounding** in `CostRollup.as_dict()`.
- **Legacy events without ``model_id`` parsed gracefully** — fall back to Sonnet default at lookup (conservative overestimate).

C3 — RBAC role admin (≈ 5 min)

Open at: <https://cra-test-alb-1950107331.us-east-1.elb.amazonaws.com/admin.html>

What to demo

- **Grant role to user** — pick user, role, scope (global / study / site / sr_review), scope_id.
- **Revoke role** — list assignments; click revoke; row deletes.
- **Effective permissions** — query `effective_permissions_for_sub(sub)` → shows the permission set the user actually carries.
- **Frontend gating** — sidebar links + welcome-page tiles disappear when the user lacks the corresponding permission.

Sanity checks

- **Three-tier scope.** `global ⊃ study ⊃ site`. Broader scope satisfies narrower checks.
- **``sr_review`` scope independent.** A `study` grant doesn't satisfy an `sr_review` check.
- **``require_permission_scoped(perm, resource_param=...)``** — resolves the resource id off the request path, runs through the right resolver, asks the RBAC module whether the caller's permissions cover it.
- **Foreign rows 404 not 403** to avoid id leaks.

C4 — Library + RAG (≈ 5 min)

Open at: <https://cra-test-alb-1950107331.us-east-1.elb.amazonaws.com/library.html>

What to demo

- **Library list** — every abstract + full-text article pulled across reviews.
- **Search box** runs `rag_search` (hybrid BM25 + dense vector via Titan v2 embeddings).
- **Upload a PDF** → parses + embeds + appears in the library.
- **Re-import lineage** — same paper across multiple reviews shows as one library entry with N citing threads.

Sanity checks

- **HNSW index** (Postgres + pgvector) for vector search.
- **PDF upload** uses pypdf for parsing.
- **Embedding dimension** = 1024 (Titan v2). Must match `models.EMBEDDING_DIM`.
- **RAG hits surface in any specialist** that has `rag_search` registered (meta_analysis, sr_protocol, general_qa).

C5 — General clinical Q&A (≈ 3 min)

Demos. Anti-hallucination validator on the general-purpose chat surface.

Steps

1. Ask a defensible question:

/ask what does a forest plot show in a meta-analysis?

→ background explanation citing methodology textbook patterns; no PMIDs, no effect sizes.

2. Try to get a fabricated effect size:

What's the pooled HR for SGLT2 inhibitors on cardiovascular mortality?

→ `_reject_clinical_synthesis` validator fires. The response refuses to quote a numeric HR and redirects to the meta-analysis specialist.

3. Try to get a fabricated PMID:

Give me five PMIDs for landmark sepsis trials.

→ validator fires. Response says it can't emit PMIDs from training data; suggests running `/search` instead.

Sanity checks

- ``_reject_clinical_synthesis` regex` blocks effect-size patterns (HR / OR / RR + CIs + p-values).
- **PMID rule** blocks any 8-digit PMID-like number that didn't come from a `search_papers` tool result in the same thread.
- **Guideline-citation regex** blocks "per [guideline] §[x.y]" claims without a tool-result source.

Appendix — All paste blocks (cheat sheet)

For when you just want every paste block in one place, in demo order.

E1 — Meta-analysis on synthetic data

Does adding a proton pump inhibitor to dual antiplatelet therapy reduce upper GI bleeding in patients who recently underwent PCI for acute coronary syndrome?

```
| Element | Value |
|---|---|
| Population | Adults post-PCI for ACS, on dual antiplatelet therapy (aspirin + P2Y12 inhibitor) |
| Intervention | PPI added to DAPT (any PPI, any dose, ≥30 days) |
| Comparator | DAPT alone or DAPT + placebo |
| Outcome (primary) | Upper GI bleeding events during follow-up |
| Study design | Randomized controlled trials |

| Study | Year | Design | PPI events | PPI total | Control events | Control total | Follow-up (mo) |
|---|---|---|---|---|---|---|---|
| Anderson et al. | 2014 | Multicenter RCT | 12 | 1200 | 28 | 1198 | 12 |
| Bekele et al. | 2016 | Single-center RCT | 4 | 320 | 9 | 318 | 6 |
| Chen et al. | 2018 | Multicenter RCT | 18 | 2200 | 41 | 2180 | 12 |
| Diaz et al. | 2019 | Pragmatic RCT | 7 | 850 | 14 | 845 | 9 |
| Eriksen et al. | 2020 | RCT | 5 | 560 | 7 | 562 | 6 |
| Faruq et al. | 2021 | Multicenter RCT | 10 | 1400 | 25 | 1395 | 12 |
| Garcia et al. | 2022 | RCT | 3 | 480 | 8 | 478 | 6 |
| Huang et al. | 2023 | Multicenter RCT | 14 | 1800 | 33 | 1810 | 12 |
```

E1 variant — high heterogeneity

```
| Study | Year | Design | PPI events | PPI total | Control events | Control total | Follow-up (mo) | Approx
RR |
|---|---|---|---|---|---|---|---|
| Anderson et al. | 2014 | Multicenter RCT | 8 | 1200 | 20 | 1198 | 12 | 0.40 |
| Bekele et al. | 2016 | Single-center RCT | 3 | 400 | 12 | 395 | 6 | 0.25 |
| Chen et al. | 2018 | Multicenter RCT | 15 | 1500 | 25 | 1495 | 12 | 0.60 |
| Diaz et al. | 2019 | Pragmatic RCT | 22 | 600 | 12 | 605 | 9 | 1.85 |
| Eriksen et al. | 2020 | RCT | 4 | 700 | 16 | 695 | 6 | 0.25 |
| Faruq et al. | 2021 | Multicenter RCT | 18 | 900 | 14 | 895 | 12 | 1.28 |
| Garcia et al. | 2022 | RCT | 5 | 800 | 19 | 795 | 6 | 0.26 |
| Huang et al. | 2023 | Multicenter RCT | 11 | 1300 | 24 | 1305 | 12 | 0.46 |
```

E2a — Protocol triggers

/protocol SGLT2 inhibitors for HF prevention in T2DM

Help me draft a PRISMA-P protocol on whether SGLT2 inhibitors reduce heart-failure hospitalization in adults with type 2 diabetes.

E2b — Search-strategy triggers

/search proton pump inhibitor + DAPT in post-PCI ACS

E2d — RoB standalone trigger

/rob Run RoB 2.0 on PMID 20925534, PMID 30873575, PMID 31091374

E4 — NMA trigger

/nma compare five direct oral anticoagulants for stroke prevention in non-valvular AF – apixaban, dabigatran, edoxaban, rivaroxaban, warfarin

E5 — IPD trigger + bundle

/ipd individual patient data meta-analysis on statins for primary prevention of cardiovascular events

Trial: ASCOT-LLA

subject_id,treatment,outcome,age,sex

S001,1,0,55,M
S002,1,1,62,M
S003,0,1,58,F
S004,0,0,49,M

Trial: WOSCOPS
subject_id,treatment,outcome,age,sex
W001,1,0,60,M
W002,0,1,67,M
W003,1,0,54,F
W004,0,0,52,F

E6 — GRADE trigger

/grade Generate a GRADE SoF for the PPI/DAPT meta-analysis

D1 — SAP drafter trigger

/sap two-arm RCT of CBT vs SSRI for depression, primary outcome HAM-D at 12 weeks

S1 — Registration trigger

/register Phase 2 SGLT2i in HFpEF, double-blind randomised parallel, primary outcome KCCQ Total Symptom Score at week 24

S2 — IRB trigger

/irb Phase 2 trial of Drug X in disease Y, US IRB, language English, reading grade 8

X1 — Protocol paste for eCRF drafter

Phase 2, single-arm, open-label study of an SGLT2 inhibitor in adults with type 2 diabetes mellitus.

Eligibility:

- Adults aged 18–75 years.
- HbA1c 7.0–10.0% at screening.
- On stable metformin ≥ 3 months prior to enrolment.
- BMI 22–40 kg/m².

Schedule of assessments:

- Visit 1 (Baseline, Week 0): consent, demographics, vital signs, fasting glucose, HbA1c, body weight, eligibility verification, study-drug dispensing.
- Visit 2 (Week 4): vital signs, fasting glucose, AE review.
- Visit 3 (Week 12): vital signs, fasting glucose, HbA1c, body weight, AE review, lab safety panel.
- Visit 4 (Week 24, primary endpoint): vital signs, fasting glucose, HbA1c, body weight, AE review, lab safety panel.
- Visit 5 (EOS, Week 28): final vital signs, AE review, treatment-emergent serious AE inventory.

Primary outcome: change in HbA1c from baseline to Week 24.

Secondary outcomes: fasting glucose, body weight, proportion with HbA1c <7.0% at Week 24.

Safety outcomes: vital signs, lab safety panel, treatment-emergent adverse events (graded per CTCAE v5).

Patient-reported outcomes (ePRO, weekly):

- Diabetes Treatment Satisfaction Questionnaire (DTSQ, 8 items).
- Self-reported hypoglycaemia events (frequency + severity).

X10 — Lab feed samples

HL7 v2 ORU^R01:

MSH|^~\&|LAB|HOSP|EDC|TRIAL|20260531120000||ORU^R01|MSG1|P|2.5
PID|||S01-001

```
OBR|1||ACC123|CBC^Complete Blood Count^L|||20260531110000
OBX|1|NM|HGB^Hemoglobin^L||14.2|g/dL|13.0-17.0|N||F
OBX|2|NM|GLUC^Glucose^L||102|mg/dL|70-100|H||F
```

CDISC LAB:

```
STUDYID■SITEID■SUBJID■ACCSNNUM■LBTESTCD■LBTEST■LBORRES■LBORRESU■LBORNRL0■LBORNRHI■LBNRIND■LBDTC
STUDY1■01■S01-001■A1■CREAT■Creatinine■0.9■mg/dL■0.7■1.3■NORMAL■2026-05-31T11:05:00
STUDY1■01■S01-001■A1■BUN■Urea Nitrogen■18■mg/dL■7■20■NORMAL■2026-05-31T11:05:00
```

FHIR R4:

```
{
  "resourceType": "Bundle",
  "type": "collection",
  "entry": [
    {
      "resource": {
        "resourceType": "Observation",
        "subject": { "reference": "Patient/S01-001" },
        "code": { "coding": [{ "code": "TSH", "display": "TSH" }] },
        "valueQuantity": { "value": 2.1, "unit": "mIU/L" },
        "referenceRange": [{ "low": { "value": 0.4 }, "high": { "value": 4.0 } }],
        "effectiveDateTime": "2026-05-31T11:05:00Z"
      }
    }
  ]
}
```

A1 — Trial-stats trigger

/trial-stats run efficacy + safety analyses on the SGLT2i SMOKE-T2DM-001 ADaM bundle, primary endpoint HbA1c at Week 24

A4 — CSR trigger

/csr Phase 2 study of Empagliflozin in adults with T2DM, lock date 2026-05-31, double-blind, FDA submission

A5 — Manuscript trigger + reviewer comments

/manuscript compose an IMRaD manuscript from my last meta-analysis, target NEJM

Reviewer 1:

- The forest plot doesn't include a leave-one-out sensitivity analysis. Please add.
- Some included studies appear to have high risk of bias. How did this affect conclusions?

Reviewer 2:

- The discussion is too long. Please cut by 30%.

A6 — Lay summary trigger

/lay-summary plain-language recruitment material at grade 6 English for parents of children with asthma – Phase 2 RCT of inhaled corticosteroid X vs standard care

A7 — Sample BibTeX

```
@article{anderson2014,
  author = {Anderson, J. and Smith, K.},
  title = {Proton Pump Inhibitor Use in ACS},
  journal = {Lancet},
  year = {2014},
  pmid = {25040175}
}
```

Slash commands

/protocol, /sr, /prisma	→ sr_protocol specialist
/search, /strategy	→ search_strategy specialist
/meta, /meta-analysis, /ma	→ meta_analysis specialist
/rob, /bias	→ risk_of_bias specialist
/nma, /network-ma	→ nma specialist
/ipd, /ipdma	→ ipd specialist
/grade, /sof, /prisma-checklist	→ grade_drafter specialist
/sap, /samplesize, /power	→ sap_drafter specialist
/register, /ctgov, /euctr, /ctis	→ registration_drafter specialist
/irb, /icf, /consent	→ irb_drafter specialist
/manuscript, /imrad	→ manuscript_drafter specialist
/csr, /e3, /study-report	→ csr_drafter specialist
/trial-stats, /efficacy, /km, /mmrm	→ trial_stats specialist
/lay, /lay-summary, /pls	→ lay_summary specialist
/general, /ask	→ general_qa specialist

Walking a customer through a phase

If you're demoing to a researcher / sponsor / institution, pick the phase that matches their pain point and run the demos in order:

- **A researcher writing a SR / meta-analysis** → Phase 01 demos E1 → E2 → E3 → E6 (≈ 35 min).
- **A guideline committee / HTA body** → Phase 01 demos E2 → E7 → E8 → E6 (≈ 30 min).
- **A trial PI preparing to submit a protocol** → Phase 02 D1 → Phase 03 S1 → S2 → S3 (≈ 35 min).
- **A site coordinator / data manager running an active trial** → Phase 04 X1 → X2 → X4 → X5 → X6 → X7 → X8 → X10 → X11 (≈ 60 min).
- **A biostatistician + medical writer at database lock** → Phase 04 X13 → Phase 05 A1 → A3 → A4 → A5 (≈ 45 min).
- **A patient-engagement / comms team** → Phase 05 A6 + Phase 01 E2 → E6 (≈ 25 min).
- **An institutional admin** → Phase 06 C1 → C2 → C3 → Phase 04 X11 → X13 (≈ 30 min).

For a 15-minute "elevator demo" pick one from each phase: E1 → D1 → S2 → X2 → A4 → C2. Shows the lifecycle end-to-end with the cross-handoff story.

Functional-test mode

Every demo doubles as an end-to-end functional test. To run the full surface as an acceptance check:

1. Open the app and sign in (see **Before you start**).
2. Walk E1 through C5 in order, executing each demo's **Steps**.
3. Confirm every **Sanity check** passes.
4. Note any **Failure modes** that fire unexpectedly.

A complete pass takes ~3.5 hours and exercises every shipped specialist, every eCRF endpoint, the SDTM/ADaM/TLF pipeline, the validation pack, the RBAC matrix, and the dashboards. Failure-mode coverage is intentional — most regressions surface here, not in the unit suite.